

VaultX AI Internship — Week 02

Task 3: Classification Tool — Evaluation Results

1. Objective

Build a classification tool that takes a customer support message as input and returns validated JSON containing four fields: category, priority, sentiment, and needs_human. The tool reuses the structured-output function (get_structured_ticket) built in Task 2, which enforces a Pydantic schema (SupportTicket) and retries automatically if the model's JSON response fails validation.

2. Schema Used

category: one of billing, technical, general, account, complaint
priority: one of low, medium, high
sentiment: one of positive, neutral, negative
needs_human: boolean (true/false)

3. Test Methodology

20 realistic support messages were authored covering a mix of billing, technical, account, complaint, and general categories, with varying priority and sentiment levels. Each message was passed to get_structured_ticket(), and the model's predicted output was compared field-by-field against a manually pre-defined expected output. A sample was marked correct only if all four fields matched exactly.

4. Results Summary

Total samples tested: 20

Fully correct (all 4 fields match): 11

Accuracy: 55.0%

Note: The internship document does not specify a minimum required accuracy for Task 3. This 55% figure represents the baseline performance of the first prompt version and is intended to be improved during Task 5 (prompt evaluation and iteration).

5. Detailed Results

#	Message	Expected (cat/pri/sent/human)	Predicted (cat/pri/sent/human)	Match
1	My payment failed twice and I'm really frustrated, p...	billing/high/negative/True	billing/high/negative/True	Yes
2	What are your business hours?	general/low/neutral/False	general/low/neutral/False	Yes
3	The app keeps crashing every time I open the camera ...	technical/high/negative/True	technical/high/negative/True	Yes
4	Can I change my email address on my account?	account/low/neutral/False	account/medium/neutral/True	No
5	I was charged twice for the same subscription this m...	billing/high/negative/True	billing/high/negative/True	Yes
6	How do I reset my password?	account/low/neutral/False	account/medium/neutral/True	No
7	This is the third time your service has failed me. I...	complaint/high/negative/True	complaint/high/negative/True	Yes
8	Just wanted to say your support team was super helpf...	general/low/positive/False	general/low/positive/False	Yes
9	The dashboard is loading very slowly since the last ...	technical/medium/negative/True	technical/medium/negative/True	Yes
10	Do you offer a student discount?	billing/low/neutral/False	general/medium/neutral/True	No
11	I can't log in, it says my account is locked.	account/high/negative/True	account/high/negative/True	Yes
12	Your product is great, but the mobile app UI could b...	general/low/positive/False	general/medium/positive/False	No
13	I need to cancel my subscription immediately, this i...	complaint/high/negative/True	billing/high/negative/True	No
14	Is there an API available for developers?	technical/low/neutral/False	technical/medium/neutral/False	No
15	My invoice shows the wrong amount, please correct it.	billing/medium/negative/True	billing/high/negative/True	No
16	How do I update my billing address?	account/low/neutral/False	billing/medium/neutral/True	No
17	The export feature is completely broken, I lost impo...	technical/high/negative/True	technical/high/negative/True	Yes
18	Thanks for fixing the bug so quickly, really appreci...	general/low/positive/False	general/low/positive/False	Yes
19	I've been on hold for 40 minutes and no one is respo...	complaint/high/negative/True	complaint/high/negative/True	Yes
20	Can you tell me the difference between the Pro and B...	general/low/neutral/False	billing/medium/neutral/True	No

6. Observations on Mismatches

Most mismatches occurred on two fields: **priority** and **needs_human**, particularly on simple FAQ-style messages (e.g. "How do I reset my password?", "Is there an API available for developers?"). The expected labels treated these as low priority and not requiring a human, but the model frequently rated them medium priority and needing human review — suggesting the prompt does not clearly define thresholds for when human escalation is or is not needed.

One clear category mismatch was "How do I update my billing address?", which the model classified as

billing, while the expected label was account (updating an address is an account/profile change, not a payment issue). This indicates the prompt could benefit from clearer category definitions distinguishing account-settings changes from payment/invoice issues.

7. Conclusion

The classification tool successfully produces validated, schema-conformant JSON output for all 20 test messages with zero validation failures, satisfying the structured-output requirement. The baseline accuracy of 55% highlights specific, identifiable weaknesses in the current prompt (ambiguous priority/needs_human thresholds and category boundary confusion) rather than a broken system. These findings will guide prompt iteration in Task 5.